

Finite Hankel Certificates for Three Somos–8 Conjectures

The Somos–8 assertions in Barry’s Conjectures 11–13,
singular specializations, and coefficient asymptotics

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19 September 2026

Result and proof status

The three parameterized bilinear Hankel identities are established by combining Hone’s published genus-two theorem with a specialization argument and twelve exact polynomial identities. The local certificates are reproduced by a standard-library Python verifier and independently checked with SymPy. The proof does not assume the conjectured recurrences to generate its data. It remains valid in denominator-cleared form at singular parameter values. For Conjecture 13, the defining continued fraction is used; a sign error in the source’s separate Catalan display is corrected explicitly. No earlier resolution of these three Somos–8 assertions was located in the focused literature search. This is not an exhaustive certification of novelty.

Research article and reproducibility package. The genus-two existence theorem is an external mathematical input, not a theorem reproved in this report.

Abstract

Barry proposed three one-parameter families of algebraic generating functions whose Hankel transforms were conjectured to satisfy explicit Somos–8 recurrences. We prove the three bilinear assertions, namely the Somos–8 parts of Conjectures 11, 12, and 13 in arXiv:2211.12637. The argument makes a finite computation logically sufficient: Hone’s genus-two Hankel theorem implies that an infinite matrix of quadratic products has rank at most four; a binomial translation puts each proposed family into the corresponding universal moment class; and four symbolic rows determine the recurrence. A polynomial-specialization argument is essential because the proposed families lie on a degenerate parameter locus of the usual continued-fraction coordinates. We give denominator-free formulas, exact nonzero rank witnesses, and executable integer-polynomial certificates. For the second family we also derive two Somos–6 specializations, a positive finite coefficient formula with a tree interpretation, and an all-orders dominant-singularity expansion of its generating-function coefficients. These asymptotics concern the moments, not their Hankel determinants.

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1 The selected problem and its provenance

A useful research target should have an unambiguous published statement, enough structure to support an all-index argument, and a feasible independent verification path. The Somos–8 conjectures in Barry’s 2022 preprint [1] meet these requirements. They concern an algebraic power series $g(x) = \sum_{n \geq 0} a_n x^n$ and its Hankel transform

$$h_n = \det(a_{i+j})_{0 \leq i, j \leq n}, \quad n \geq 0. \quad (1.1)$$

The striking assertion is not integrality of h_n , which follows directly when the a_n are integers. It is that the determinants obey a fixed short bilinear recurrence with explicitly conjectured rational coefficients.

Our main target is Conjecture 12. Its generating function is the unique formal power-series solution with constant term one of

$$g(x) = \frac{1}{1 - x - \frac{x^2}{1 - rx} - x^3 g(x)}. \quad (1.2)$$

The proposed Hankel recurrence has order eight. The proof developed for this family also proves the adjacent Conjectures 11 and 13 with no new structural input, so all three are treated together.

There are important distinctions in the literature. Wang and Zhang [3] prove the four Somos–4 conjectures in Barry’s paper, not the three Somos–8 assertions studied here. Hone’s earlier work [2] provides the general genus-two theorem that makes our proof possible, but its statement does not list Barry’s later one-parameter coefficient formulas. The July 2026 paper of Liu, Wang, and Zhang [4], despite its similar title, concerns four conjectures about a *central transform*. The September 2026 note of Scheuerle [5] studies inverse realization of prescribed Somos–4 Hankel data. These are different problems.

The search leading to this report checked the original preprint, the cited Somos–4 resolution, Hone’s theorem and its moment parameterization, and recent papers surfaced by searches for Barry’s Hankel conjectures. No prior proof of the three precise Somos–8 formulas was located. This supports selecting them as conjectural targets, but it does not prove that no such proof exists elsewhere. The mathematical claims below are proofs of the stated identities; historical priority is a separate question.

1.1 What is established, and what is imported

The only substantial structural input is Hone’s genus-two theorem [2, Theorem 5.4]. Its use is explicit. We prove the translation, the polynomial-specialization step, the finite-certification lemma, the parameter substitutions, the exceptional-parameter consequences, and the coefficient formulas and asymptotics. The finite identities needed to identify the recurrence are evaluated exactly in $\mathbb{Z}[r]$, using two independent determinant implementations.

This is therefore not merely a numerical confirmation, and it is not a self-contained rederivation of genus-two integrability. It is a proof by reduction to a published theorem, with all family-specific algebra made reproducible. We do not claim a proof of the other conjectures in Barry’s paper, nor a classification of all algebraic generating functions with Somos Hankel transforms.

2 Precise statements of the three identities

2.1 Uniform notation

Let r be an indeterminate and set $B(x) = 1 - rx$. Define

$$A_{11}(x) = 1 - (r + 1)x - x^2 + rx^3, \quad (2.1)$$

$$A_{12}(x) = 1 - (r + 1)x + (r - 1)x^2, \quad (2.2)$$

$$A_{13}(x) = 1 - (r + 1)x + (r - 2)x^2 + rx^3. \quad (2.3)$$

For $k \in \{11, 12, 13\}$, define $g^{(k)}(x) \in \mathbb{Z}[r][[x]]$ by

$$A_k(x)g^{(k)}(x) = B(x) + x^3B(x)g^{(k)}(x)^2, \quad g^{(k)}(0) = 1. \quad (2.4)$$

Write its coefficients as $a_n^{(k)}$, and use (1.1) for $h_n^{(k)}$. We also define the empty determinant

$$h_{-1}^{(k)} = 1. \quad (2.5)$$

The extra index allows the bilinear identity to start at $n = 7$, one step before the customary recurrence written solely with nonnegative indices.

Dividing (2.4) by $B(x)g^{(k)}(x)$ yields respectively

$$g^{(11)}(x) = \frac{1}{1 - \frac{x}{1 - rx} - x^2 - x^3g^{(11)}(x)}, \quad (2.6)$$

$$g^{(12)}(x) = \frac{1}{1 - x - \frac{x^2}{1 - rx} - x^3g^{(12)}(x)}, \quad (2.7)$$

$$g^{(13)}(x) = \frac{1}{1 - \frac{x(1 - (r - 1)x)}{1 - rx} - x^2 - x^3g^{(13)}(x)}. \quad (2.8)$$

These are the defining continued fractions in Barry's Conjectures 11–13.

Remark 2.1 (A source correction, not a counterexample). In the separate Catalan generating-function display for Conjecture 13, Barry prints $1 + (r + 1)x + (r - 2)x^2 + rx^3$. The defining continued fraction (2.8) instead gives (2.3), with a minus sign before $(r + 1)x$. This follows by multiplying its denominator by $1 - rx$. We use the continued-fraction definition throughout. The claimed Somos parameters require no change. Our result establishes that substantive recurrence claim, not the inconsistent pair of generating-function displays read literally.

2.2 Denominator-free coefficient tables

The following polynomials specify the answer. The subscript on $P_{k,j}$ is the family number followed by the position of the quadratic product. Using a common denominator avoids spurious exclusions at its roots.

For the family $k = 12$, the simplest of the three, let

$$D_{12} = r^2 - 4r + 3 = (r - 1)(r - 3), \quad (2.9)$$

$$P_{12,1} = r^4 - 11r^3 + 26r^2 - 16r - 5, \quad (2.10)$$

$$P_{12,2} = 2r^5 - 19r^4 + 40r^3 - 13r^2 - 5r, \quad (2.11)$$

$$P_{12,3} = (r - 1)(3r^6 - 12r^5 + 15r^4 + 25r^3 - 62r^2 - 36r - 5), \quad (2.12)$$

$$P_{12,4} = r^9 - 8r^8 + 26r^7 - 43r^6 + 40r^5 - 17r^4 - 23r^3 + 27r^2 + 19r + 3. \quad (2.13)$$

For $k = 11$, put

$$D_{11} = r^4 - 2r^3 + 8r^2 + 2r - 9 = (r^2 - 1)(r^2 - 2r + 9), \quad (2.14)$$

$$P_{11,1} = r^8 - 8r^7 + 21r^6 - 40r^5 + 35r^4 - 24r^3 + 71r^2 + 8r, \quad (2.15)$$

$$P_{11,2} = 8(r^9 - 6r^8 + 17r^7 - 30r^6 + 15r^5 - 14r^4 - r^3 - 14r^2), \quad (2.16)$$

$$P_{11,3} = 8(r - 1)(r^{10} - 2r^8 + 29r^7 - 32r^6 + 39r^5 + 18r^4 + 11r^3 - r^2 + r), \quad (2.17)$$

$$P_{11,4} = 2r^{13} - 13r^{12} + 48r^{11} - 85r^{10} + 83r^9 - 11r^8 + 124r^7 - 454r^6 + 364r^5 - 263r^4 - 84r^3 - 189r^2 - 25r - 9. \quad (2.18)$$

Finally, for $k = 13$, put

$$D_{13} = 2(r^3 - 3r^2 - 5r + 7) = 2(r - 1)(r^2 - 2r - 7), \quad (2.19)$$

$$P_{13,1} = -(r^7 - 8r^6 + 25r^5 - 20r^4 - 37r^3 + 75r + 28), \quad (2.20)$$

$$P_{13,2} = (r + 1)(r^8 - 11r^7 + 47r^6 - 83r^5 + 17r^4 + 71r^3 + 45r^2 - 169r + 210), \quad (2.21)$$

$$P_{13,3} = (r^2 - 1)(3r^8 - 29r^7 + 115r^6 - 225r^5 + 181r^4 + 105r^3 - 255r^2 - 235r + 84), \quad (2.22)$$

$$P_{13,4} = -(r^{10} - 17r^9 + 96r^8 - 212r^7 + 54r^6 + 594r^5 - 796r^4 - 36r^3 + 721r^2 - 329r - 588). \quad (2.23)$$

Theorem 2.2 (Three Somos–8 identities). *For every $k \in \{11, 12, 13\}$ and every integer $n \geq 7$, the identity*

$$D_k(r)h_n^{(k)}h_{n-8}^{(k)} = \sum_{j=1}^4 P_{k,j}(r)h_{n-j}^{(k)}h_{n-8+j}^{(k)} \quad (2.24)$$

holds in $\mathbb{Z}[r]$. In particular, for a complex parameter r with $D_k(r) \neq 0$, these Hankel determinants satisfy the Somos–8 bilinear relation with coefficients

$$(\alpha, \beta, \gamma, \delta) = D_k(r)^{-1}(P_{k,1}, P_{k,2}, P_{k,3}, P_{k,4}). \quad (2.25)$$

These are precisely the Somos–8 coefficient formulas in Barry's Conjectures 11, 12, and 13, interpreted using their defining continued fractions. At roots of D_k , the polynomial relation (2.24) still holds.

The proof is given in Sections 4–7. For $k = 11$, the factor $r - 1$ in $P_{11,3}$ explains why the source uses a smaller denominator for γ ; for $k = 12$, the analogous factor accounts for its denominator $r - 3$. Thus no coefficients have been fitted approximately or altered to obtain the theorem.

Corollary 2.3 (Integrality and arbitrary specializations). *For every integer r , all moments and Hankel determinants are integers. More generally, (2.24) specializes to every commutative ring with identity by sending r to any element of that ring.*

Proof. The coefficient recursion from (2.4) uses only addition and multiplication in $\mathbb{Z}[r]$; see (3.2) below. Determinants therefore also belong to $\mathbb{Z}[r]$. Apply a ring homomorphism to the polynomial identity in Theorem 2.2. $\square$

This last statement does not apply a genus-two analytic theorem directly in positive characteristic. It specializes an integral polynomial identity already proved in characteristic zero.

3 Formal power series and exact moment generation

Write $A_k(x) = 1 + A_1x + A_2x^2 + A_3x^3$, with $A_3 = 0$ when $k = 12$. In all three cases $A_1 = -(r + 1)$. Set $a_n = 0$ for negative n , and

$$c_m = \begin{cases} \sum_{i=0}^m a_i a_{m-i}, & m \geq 0, \\ 0, & m < 0. \end{cases} \quad (3.1)$$

Equating coefficients in (2.4) gives $a_0 = 1$ and

$$a_n = -A_1 a_{n-1} - A_2 a_{n-2} - A_3 a_{n-3} + c_{n-3} - r c_{n-4} - r \mathbf{1}_{n=1}, \quad n \geq 1. \quad (3.2)$$

Every term on the right is already known at stage n . This proves both existence and uniqueness of the formal series, as well as the asserted integral-polynomial coefficient property. The first coefficients are

$$a_0 = 1, \quad a_1 = 1, \quad a_2 = \begin{cases} r + 2, & k = 11, \\ 2, & k = 12, \\ 3, & k = 13. \end{cases} \quad (3.3)$$

If $C(z) = \sum_{m \geq 0} C_m z^m$ is the Catalan series, characterized by $C(z) = 1 + zC(z)^2$, then

$$g^{(k)}(x) = \frac{B(x)}{A_k(x)} C\left(\frac{x^3 B(x)^2}{A_k(x)^2}\right). \quad (3.4)$$

This follows by substitution, or from the branch

$$g^{(k)}(x) = \frac{A_k(x) - \sqrt{A_k(x)^2 - 4x^3 B(x)^2}}{2x^3 B(x)}, \quad (3.5)$$

where the square root has constant term one. The apparent singularity at $x = 0$ in (3.5) is removable as a formal series.

It is essential that our computation proceeds from (3.2) and then evaluates determinants. Generating h_n from the proposed Somos recurrence and comparing those values with the same recurrence would be circular. No such step is used in the verifier.

For orientation, the first few Hankel polynomials for $k = 12$ are

$$\begin{aligned} h_{-1} = h_0 = h_1 &= 1, & h_2 &= -2r, \\ h_3 &= -r^4 + 2r^3 - r^2 - 4r - 1, \\ h_4 &= -r^6 + 4r^5 - 5r^4 + r^3 - 6r^2 - 5r - 1, \\ h_5 &= r^7 - 5r^6 + r^5 + 12r^4 + 12r^3 - 7r^2 - 6r - 1. \end{aligned} \quad (3.6)$$

The relatively small degrees of these polynomials make exact family-level certification much less expensive than a general seven-parameter calculation.

4 The structural input: a rank-four theorem

4.1 A universal genus-two moment class

Introduce seven algebraically independent parameters $u, v, b_2, b_3, c_2, c_3, c_4$, and moments $s_0 = 1, s_1 = u, s_2 = v$. For $m \geq 3$, define

$$\begin{aligned} s_m &= b_2 s_{m-2} + b_3 s_{m-3} + c_2 \sum_{i=0}^{m-2} s_i s_{m-2-i} \\ &\quad + c_3 \sum_{i=0}^{m-3} s_i s_{m-3-i} + c_4 \sum_{i=0}^{m-4} s_i s_{m-4-i}, \end{aligned} \quad (4.1)$$

with an empty sum equal to zero. Let

$$H_N = \det(s_{i+j})_{0 \leq i, j < N}, \quad H_0 = 1. \quad (4.2)$$

Notice the size convention: H_N is an N -by- N determinant, whereas h_n is an $(n+1)$ -by- $(n+1)$ determinant.

Theorem 4.1 (Hone’s genus-two theorem, the form used here). *For the generic moment sequence (4.1), the determinants (4.2) satisfy a nontrivial relation*

$$q_0 H_{N+8} H_N + q_1 H_{N+7} H_{N+1} + q_2 H_{N+6} H_{N+2} + q_3 H_{N+5} H_{N+3} + q_4 H_{N+4}^2 = 0, \quad (4.3)$$

where the coefficients are independent of N .

This is the input from [2, Theorem 5.4, equation (4.22), and the seven-parameter formulation in its proof]. We require existence and index-independence of the relation, not the large general formulas for the q_j . In the generic setting, the relation holds for $N \geq 0$.

For clarity, the link from (4.1) to Hone’s curve coordinates can be exhibited directly. Put

$$G(X) = \sum_{m \geq 0} s_m X^{-m-1}, \quad (4.4)$$

and define

$$\begin{aligned} P(X) &= X^3 - b_2 X - b_3, \\ S(X) &= c_2 X^2 + c_3 X + c_4, \\ T(X) &= X^2 + uX + v - b_2 - c_2. \end{aligned} \quad (4.5)$$

The coefficient recursion is equivalent to

$$P(X)G(X) - S(X)G(X)^2 = T(X). \quad (4.6)$$

Indeed, its polynomial part specifies s_0, s_1, s_2 , while each negative power of X gives (4.1). Setting

$$Y = P - 2SG \quad (4.7)$$

yields

$$Y^2 = P^2 - 4ST. \quad (4.8)$$

Generically this is a nonsingular sextic model of a genus-two curve. Moreover, $Q_0 = -2S$ and $Q_1 = 2T$ put (4.6) into Hone's equation $PG + Q_0G^2/2 = Q_1/2$. The polynomial part of the square root at infinity is

$$X^3 - (b_2 + 2c_2)X - (b_3 + 2c_2u + 2c_3). \quad (4.9)$$

Thus the seven parameters in (4.1) are the generic moment parameters of the cited theorem, not an additional unproved generalization of it.

4.2 Why specialization needs an argument

All moments in (4.1), and hence all H_N , are polynomials in the seven parameters. Consider the infinite matrix whose row indexed by $N \geq 0$ is

$$(H_{N+8}H_N, H_{N+7}H_{N+1}, H_{N+6}H_{N+2}, H_{N+5}H_{N+3}, H_{N+4}^2). \quad (4.10)$$

Theorem 4.1 implies that its five columns are linearly dependent over the generic parameter field.

Lemma 4.2 (Polynomial rank specialization). *Every five-by-five minor of (4.10), using any five row indices, is the zero polynomial. Consequently, after any specialization of the seven parameters into a characteristic-zero field, the specialized matrix has rank at most four.*

Proof. Fix any five row indices. Their minor is a polynomial in the seven parameters. It is zero in the generic field by Theorem 4.1, so it is the zero polynomial. Equivalently, it vanishes on the generic set where the theorem applies and hence identically. Substitution in a zero polynomial preserves zero. Since the chosen rows were arbitrary, all five-by-five minors vanish after specialization. $\square$

The lemma concerns *arbitrary* row selections, not merely five consecutive rows. The existence of a single index-independent relation in Hone's theorem is what supplies this stronger conclusion. Also, the rank cannot rise when specializing polynomial data, although it can drop.

This step matters here because the desired families will have $c_2 = 0$. In the direct continued-fraction chart, Q_0 then loses degree and some coordinate formulas divide by c_2 . Substituting zero into those rational formulas would be invalid. Specializing the polynomial minors instead is valid and is sufficient for the proof.

5 Binomial translation and the three parameter substitutions

5.1 Hankel invariance under translating moments

Lemma 5.1 (Binomial translation). *For any sequence (a_n) and scalar c , define*

$$s_n = \sum_{j=0}^n \binom{n}{j} (-c)^{n-j} a_j. \quad (5.1)$$

Then for every $N \geq 0$,

$$\det(s_{i+j})_{0 \leq i, j < N} = \det(a_{i+j})_{0 \leq i, j < N}. \quad (5.2)$$

In Laurent-series notation, if $F(X) = \sum_{n \geq 0} a_n X^{-n-1}$, then

$$F(X+c) = \sum_{n \geq 0} s_n X^{-n-1}. \quad (5.3)$$

Proof. Let $\mathcal{L}$ be the linear functional on polynomials with $\mathcal{L}(X^n) = a_n$. Then $s_n = \mathcal{L}((X-c)^n)$. The change of basis from $1, X, \dots, X^{N-1}$ to $1, X-c, \dots, (X-c)^{N-1}$ is lower triangular with diagonal entries one. If its matrix is U , the new moment matrix is $U(a_{i+j})U^\top$, so its determinant is unchanged. For the Laurent identity, expand each $(X+c)^{-j-1}$ using the binomial series and collect the coefficient of X^{-n-1} ; it is exactly (5.1). $\square$

5.2 Removing the quadratic term in the cubic

For a fixed family, suppress its superscript. Put

$$F(X) = X^{-1}g(X^{-1}), \quad \mathcal{A}(X) = X^3A(X^{-1}) = X^3 - (r+1)X^2 + A_2X + A_3. \quad (5.4)$$

Substituting into (2.4) and multiplying by X^2 gives

$$\mathcal{A}(X)F(X) - (X-r)F(X)^2 = X(X-r). \quad (5.5)$$

The powers of X in this step are important: $g(X^{-1}) = XF(X)$, not $F(X)$.

Set

$$c = \frac{r+1}{3}, \quad p = A_2 - 3c^2, \quad q = A_3 + A_2c - 2c^3, \quad (5.6)$$

and let $G(X) = F(X+c)$. Since $\mathcal{A}(X+c) = X^3 + pX + q$, equation (5.5) becomes

$$(X^3 + pX + q)G - (X+c-r)G^2 = (X+c)(X+c-r). \quad (5.7)$$

Its coefficients at negative powers of X give, for $m \geq 3$,

$$s_m = -ps_{m-2} - qs_{m-3} + \sum_{i=0}^{m-3} s_i s_{m-3-i} + (c-r) \sum_{i=0}^{m-4} s_i s_{m-4-i}. \quad (5.8)$$

The initial data are

$$s_0 = 1, \quad s_1 = 1 - c, \quad s_2 = a_2 - 2c + c^2. \quad (5.9)$$

Thus the required substitution into (4.1) is

$$(u, v, b_2, b_3, c_2, c_3, c_4) = (1 - c, a_2 - 2c + c^2, -p, -q, 0, 1, c - r). \quad (5.10)$$

The only family-dependent input is

Family	A_2	A_3	a_2
11	-1	r	$r+2$
12	$r-1$	0	2
13	$r-2$	r	3

By Lemma 5.1, the translated sequence has the same Hankel determinants. Lemma 4.2, applied over $\mathbb{Q}(r)$, therefore proves that the infinite matrix with rows

$$(h_n h_{n-8}, h_{n-1} h_{n-7}, h_{n-2} h_{n-6}, h_{n-3} h_{n-5}, h_{n-4}^2), \quad n \geq 7, \quad (5.11)$$

has rank at most four for each of the three families. We have established an all-index structural bound before checking any of the conjectured coefficient formulas.

6 Why four exact rows complete the proof

Lemma 6.1 (Finite certification of a fixed recurrence). *Let K be a field and $h_{-1}, h_0, \dots$ a sequence in K . For $n \geq 7$, set*

$$y_n = h_n h_{n-8}, \quad v_n = (h_{n-1} h_{n-7}, h_{n-2} h_{n-6}, h_{n-3} h_{n-5}, h_{n-4}^2). \quad (6.1)$$

Assume that the matrix with rows (y_n, v_n) has rank at most four. Choose four row indices $n_1, \dots, n_4$, and suppose that the matrix $M = (v_{n_i})_{i=1}^4$ is invertible. If $D, P_1, \dots, P_4 \in K$ satisfy

$$D y_{n_i} = v_{n_i} (P_1 \ P_2 \ P_3 \ P_4)^\top \quad (1 \leq i \leq 4), \quad (6.2)$$

then the same identity holds for every $n \geq 7$.

Proof. Take the five-by-five matrix with columns (v, y) , using the four chosen rows and an arbitrary fifth row n . Its determinant is zero by the rank hypothesis. Replace its last column by D times that column minus $\sum_{j=1}^4 P_j$ times the corresponding first four columns. The new determinant is still zero. By (6.2), its first four entries in the last column are zero. Expanding on that column yields

$$\det(M) (D y_n - v_n P) = 0. \quad (6.3)$$

Invertibility of M gives the conclusion. A repeated fifth row causes no difficulty; the desired identity for the original four rows was already assumed. $\square$

The rank hypothesis is indispensable. For an arbitrary sequence, four or four million initial recurrence checks do not prove the next check. Here the rank theorem bounds the entire infinite family of possible quadratic rows in advance. The finite computation only identifies the one relation already forced to exist.

7 Exact certificates and completion of the main proof

We use the four rows $n = 7, 8, 9, 10$. They require only $h_{-1}, h_0, \dots, h_{10}$, and thus only $a_0, a_1, \dots, a_{20}$. For each family, (3.2) computes these as integer polynomials, and direct determinants give the needed Hankel polynomials.

Define the residual polynomials

$$E_{k,n}(r) = D_k h_n h_{n-8} - \sum_{j=1}^4 P_{k,j} h_{n-j} h_{n-8+j}. \quad (7.1)$$

The exact computations yield

Family	$E_{k,7}$	$E_{k,8}$	$E_{k,9}$	$E_{k,10}$
11	0	0	0	0
12	0	0	0	0
13	0	0	0	0

Here each zero means the *zero polynomial in $\mathbb{Z}[r]$* , not a floating-point value and not an interpolation guess.

The polynomial degrees involved are modest:

Family / n	Degree of $h_n(r)$										
	0	1	2	3	4	5	6	7	8	9	10
11	0	1	3	5	7	11	15	19	24	30	36
12	0	0	1	4	6	7	10	15	19	22	27
13	0	0	2	3	6	8	11	15	18	23	27

The actual coefficient arrays, including every polynomial needed to rerun the verification, are recorded in `data/certificates.json`.

To prove that the four product rows are independent over $\mathbb{Q}(r)$, one nonzero specialization suffices. At $r = 2$, the respective four-by-four determinants are

$$\begin{aligned}
 \det M_{11}(2) &= 844298284526087295172008083616, \\
 \det M_{12}(2) &= -1967013462528657296, \\
 \det M_{13}(2) &= -83222140602851259464.
 \end{aligned} \tag{7.2}$$

In particular none of the three determinant polynomials is identically zero. For the central family, the entire witness is small enough to display:

$$M_{12}(2) = \begin{pmatrix} 2512 & 87 & 172 & 169 \\ 22041 & -10048 & -1131 & 1849 \\ -719780 & -286533 & -108016 & 7569 \\ 89140155 & -7737635 & 1917567 & 6310144 \end{pmatrix}. \tag{7.3}$$

The other two matrices are included in Appendix A and the JSON certificate.

Proof of Theorem 2.2. Fix one of the three families and work first over $K = \mathbb{Q}(r)$. Section 5 proves that the matrix (5.11) has rank at most four, including the specialization $c_2 = 0$. The appropriate nonzero value in (7.2) proves that the four-row product matrix is invertible over K . The four zero polynomials in the certificate table are exactly (6.2). Lemma 6.1 therefore gives (2.24) for every $n \geq 7$ as an identity in $\mathbb{Q}(r)$. Both sides are in $\mathbb{Z}[r]$, so their difference is the zero polynomial in $\mathbb{Z}[r]$. This proves the denominator-free statement at every parameter value. Division by $D_k(r)$ where nonzero gives (2.25) and the proposed Somos–8 coefficients. $\square$

7.1 How the finite algebra is verified

The primary verifier uses only the Python standard library. Integer polynomials are stored as coefficient tuples in ascending degree. Addition and multiplication are elementary operations, and every polynomial division is checked for an exactly zero remainder.

Determinants are evaluated by fraction-free Bareiss elimination with row pivoting. If p is the current pivot and p_{old} the previous pivot, an elimination update is

$$a'_{ij} = \frac{p a_{ij} - a_{ik} a_{kj}}{p_{\text{old}}}. \quad (7.4)$$

The previous pivot is initialized to one. Divisions in this algorithm are exact; the implementation raises an exception if either an integer remainder or a polynomial remainder occurs. A row exchange is accounted for in the determinant sign. An identically zero pivot column returns a zero determinant rather than dividing by zero.

The optional independent check constructs the moments afresh in SymPy's $\mathbb{Z}[r]$ domain and evaluates all the Hankel determinants using its `DomainMatrix.det` implementation. In the execution accompanying this report, all 21 moments and all 12 determinants, including the empty one, agree for each family using SymPy 1.14.0.

A further exact audit evaluates the denominator-cleared recurrence for all three families, each integer $r = -5, -4, \dots, 7$, and each $n = 7, 8, \dots, 28$. All

$$3 \cdot 13 \cdot 22 = 858 \quad (7.5)$$

checks pass, including roots of the denominators. These are useful tests against implementation errors, but the all-index proof is the rank argument plus the polynomial certificates, not the number 858.

8 Singular parameters, zeros, and lower-order relations

8.1 Bilinear identity versus forward division

Even when $D_k(r) \neq 0$, a forward update

$$h_n = \frac{\sum_{j=1}^4 P_{k,j}(r) h_{n-j} h_{n-8+j}}{D_k(r) h_{n-8}} \quad (8.1)$$

is only legal when $h_{n-8} \neq 0$. The polynomial statement (2.24) needs neither this assumption nor nonvanishing of other Hankel determinants. The moment recursion and determinant definition always select a sequence, even when a quotient recurrence does not uniquely continue it.

For example, at $r = 0$, the families 11 and 12 have the same generating function. Their Hankel determinants begin

$$1, 1, 0, -1, -1, -1, 0, 1, 1, 1, 0, -1, -1, -1, \dots \quad (8.2)$$

The zero terms are already enough to show why unrestricted forward division is not the right theorem. In this same specialization, family 11 yields coefficients $(0, 0, 0, 1)$, whereas family 12 yields $(-5/3, 0, 5/3, 1)$. Both identities hold: uniqueness of the coefficient vector over $\mathbb{Q}(r)$ does not imply uniqueness after a special rank drop. We do not infer a global periodicity theorem here merely from the displayed initial segment.

8.2 Two explicit Somos–6 consequences of Conjecture 12

At $r = 1$, the denominator D_{12} vanishes but the numerator vector is

$$(P_{12,1}, P_{12,2}, P_{12,3}, P_{12,4}) = (-5, 5, 0, 25). \quad (8.3)$$

At $r = 3$, it is

$$(-35, -105, 980, 735). \quad (8.4)$$

After moving the outermost surviving product to the left and replacing $n - 1$ by m , Theorem 2.2 gives the following.

Corollary 8.1 (Order reduction at the exceptional parameters). *For the family $k = 12$, the following identities hold for every $m \geq 6$:*

$$r = 1 : \quad h_m h_{m-6} = h_{m-1} h_{m-5} + 5h_{m-3}^2, \quad (8.5)$$

$$r = 3 : \quad h_m h_{m-6} = -3h_{m-1} h_{m-5} + 28h_{m-2} h_{m-4} + 21h_{m-3}^2. \quad (8.6)$$

These are genuine denominator-free consequences, not limits of an ill-defined quotient. Their initial values provide useful direct examples:

	h_0	h_1	h_2	h_3	h_4	h_5	h_6
$r = 1$	1	1	-2	-5	-13	7	132
$r = 3$	1	1	-6	-49	-205	-1	84864

For instance, the first step of (8.6) reads $84864 = 3 + 34440 + 50421$.

The denominator roots of the other two families are also transparent from (2.14) and (2.19). We assert the specialized polynomial identities there, but do not claim to classify every further reduction or every vanishing numerator configuration.

8.3 Two nonsingular integer examples

For family 12 at $r = 2$, the recurrence becomes

$$h_n h_{n-8} = 5h_{n-1} h_{n-7} - 18h_{n-2} h_{n-6} + 77h_{n-3} h_{n-5} - 13h_{n-4}^2. \quad (8.7)$$

The sequence, from h_0 , starts

$$1, 1, -4, -13, -43, 87, 2512, 22041, 179945, -6856935, -162649748, \dots \quad (8.8)$$

At $r = 4$, the parameters are

$$(\alpha, \beta, \gamma, \delta) = \left(-\frac{101}{3}, -\frac{484}{3}, 4299, \frac{23359}{3} \right). \quad (8.9)$$

This also proves the specific Somos-8 assertion in Barry's Example 14: its denominator uses $x(1 - 3x)/(1 - 4x) = x + x^2/(1 - 4x)$, so its generating function is precisely our family 12 at $r = 4$. The relevant moments and Hankel values can be regenerated without using that assertion; files for $r = 0, 1, 2, 3, 4$ are included in the package.

9 A positive coefficient formula and a tree model

This section concerns the moments $a_n^{(12)}$, not the determinant sequence. In the second family, define

$$L(x) = x + \frac{x^2}{1 - rx}. \quad (9.1)$$

Equation (1.2) is equivalent to

$$g(x) = 1 + L(x)g(x) + x^3g(x)^2. \quad (9.2)$$

Thus for a nonnegative integer r , a_n counts rooted ordered trees with the following decorations. A leaf has size zero. A unary vertex is of one of two types: a short unary vertex of size one, or a long unary vertex of size two carrying an additional possibly empty word on an r -letter alphabet, each letter contributing one to the size. A binary vertex has two ordered children and size three. The size of a decorated tree is the sum of its vertex and word sizes.

There are finitely many such trees of each size because every non-leaf vertex has positive size. The root decomposition is exactly (9.2); hence this interpretation follows from formal uniqueness. In particular $a_n(r) \in \mathbb{N}[r]$, even though the compact recurrence (3.2) contains subtraction.

For nonnegative integers j, ℓ , put

$$W(j, \ell) = \begin{cases} 1, & j = \ell = 0, \\ 0, & j = 0, \ell > 0, \\ \binom{j + \ell - 1}{\ell}, & j \geq 1. \end{cases} \quad (9.3)$$

Proposition 9.1 (Finite positive sum). *For every $n \geq 0$,*

$$a_n^{(12)}(r) = \sum_{\substack{k, q, j, \ell \geq 0 \\ j \leq q \\ 3k + q + j + \ell = n}} C_k \binom{2k + q}{q} \binom{q}{j} W(j, \ell) r^\ell, \quad C_k = \frac{1}{k + 1} \binom{2k}{k}. \quad (9.4)$$

The formula is a polynomial identity, so it also holds for negative or complex r , although the counting interpretation then no longer applies.

Proof. Solving (9.2) in Catalan form gives

$$g(x) = \sum_{k \geq 0} C_k x^{3k} (1 - L(x))^{-2k-1}. \quad (9.5)$$

Expand

$$(1 - L)^{-2k-1} = \sum_{q \geq 0} \binom{2k + q}{q} L^q, \quad L^q = \sum_{j=0}^q \binom{q}{j} x^{q+j} (1 - rx)^{-j}. \quad (9.6)$$

The coefficient of x^ℓ in the last factor is $W(j, \ell) r^\ell$. Collecting all contributions to x^n gives (9.4).

Alternatively, take a full binary tree with k binary vertices, in C_k ways. It has $2k + 1$ vertices at which chains of unary vertices can be inserted, including above its root. Distributing q unary vertices among these positions gives $\binom{2k+q}{q}$ choices. Choose j of the q unary vertices to be long. Distribute a total of ℓ additional letters among their words in $W(j, \ell)$ ways and color the letters in r^ℓ ways. The total size is $3k + q + j + \ell$, exactly as in the sum. $\square$

The generating function also gives a manifestly nonnegative recursion:

$$a_n = a_{n-1} + \sum_{j=0}^{n-2} r^j a_{n-2-j} + \sum_{i=0}^{n-3} a_i a_{n-3-i}, \quad n \geq 1, \quad (9.7)$$

with negative-index terms and empty sums zero. This provides a useful independent way to check the compact moment recursion.

10 Dominant-singularity asymptotics for the second family

Fix a real $r \geq 0$ throughout this section. Set

$$\Delta(x) = (1 - L(x))^2 - 4x^3, \quad (10.1)$$

so that

$$g(x) = \frac{1 - L(x) - \sqrt{\Delta(x)}}{2x^3}. \quad (10.2)$$

We give the full inverse-power expansion associated with the dominant singularity. This is not an asymptotic theorem for the Hankel determinants, and it is not a claim to include exponentially smaller contributions from every other algebraic singularity.

Lemma 10.1 (Unique dominant singularity). *There is a unique positive number ρ satisfying*

$$\rho + \frac{\rho^2}{1 - r\rho} + 2\rho^{3/2} = 1. \quad (10.3)$$

It lies in $(0, 1/r)$ when $r > 0$, and in $(0, 1)$ when $r = 0$. The function g has no singularity in $|x| < \rho$, and ρ is its only singularity on $|x| = \rho$. Moreover,

$$\Delta'(\rho) = -4\rho^{3/2}L'(\rho) - 12\rho^2 < 0. \quad (10.4)$$

Proof. The left side of (10.3) is strictly increasing from zero on the stated positive interval and exceeds one before the upper endpoint. This proves existence and uniqueness. The equality gives $1 - L(\rho) = 2\rho^{3/2}$, hence $\Delta(\rho) = 0$ and (10.4).

For $|z| < \rho$, the nonnegative coefficients of L imply $L(|z|) + 2|z|^{3/2} < 1$. A zero of $\Delta(z)$ would give a complex number w with $w^2 = z^3$ and $1 = L(z) \pm 2w$, contradicting

$$1 \leq |L(z)| + 2|z|^{3/2} \leq L(|z|) + 2|z|^{3/2} < 1.$$

Thus the square-root branch extends analytically inside that disk, and its apparent singularity at zero is removable.

If $|z| = \rho$ and $\Delta(z) = 0$, equality must hold throughout the same triangle inequalities. In particular $|L(z)| = L(\rho)$. The coefficients of both x and x^2 in L are positive; equality forces these two summands to have the same argument. Writing $z = \rho e^{i\theta}$, this requires $e^{i\theta} = e^{2i\theta}$, so $z = \rho$. The rational coefficients have no pole on the disk because $r\rho < 1$. Finally, the simple zero (10.4) gives a nonzero square-root term, so ρ really is a singularity. $\square$

Theorem 10.2 (Leading term and first correction). *Define*

$$\lambda = \frac{\sqrt{-\rho\Delta'(\rho)}}{2\rho^3}, \quad C = \frac{\lambda}{2\sqrt{\pi}}, \quad c_1 = -\frac{33}{8} + \frac{3\rho\Delta''(\rho)}{8\Delta'(\rho)}. \quad (10.5)$$

Then, as $n \rightarrow \infty$,

$$a_n^{(12)}(r) = C\rho^{-n}n^{-3/2} \left(1 + \frac{c_1}{n} + O(n^{-2}) \right). \quad (10.6)$$

The estimate is for each fixed $r \geq 0$; no uniformity in an unbounded range of r is asserted.

Proof. Set $t = 1 - x/\rho$. The nonanalytic part of (10.2) is

$$-\lambda t^{1/2} R(t), \quad R(t) = (1-t)^{-3} \sqrt{\frac{\Delta(\rho(1-t))}{-\rho\Delta'(\rho)t}}. \quad (10.7)$$

The quotient under the square root is analytic at zero with value one. Therefore R is analytic there, $R(0) = 1$, and

$$R(t) = 1 + R_1 t + O(t^2), \quad R_1 = 3 - \frac{\rho\Delta''(\rho)}{4\Delta'(\rho)}. \quad (10.8)$$

The algebraic function has a suitable indented complex neighborhood of its unique dominant singularity, by Lemma 10.1; the other zeros and poles are separated from it. Singularity transfer [6] applies to these half-integer terms.

For nonintegral a , the binomial coefficient identity is

$$[x^n](1 - x/\rho)^a = \rho^{-n} \frac{\Gamma(n-a)}{\Gamma(-a)\Gamma(n+1)}. \quad (10.9)$$

For $a = 1/2$, its gamma ratio is $n^{-3/2}(1 + 3/(8n) + O(n^{-2}))$, and $\Gamma(-1/2) = -2\sqrt{\pi}$. For $a = 3/2$, the relative contribution to the first correction is $-3R_1/(2n)$. Hence the correction is $3/8 - 3R_1/2$, which simplifies to (10.5). The analytic part does not contribute to these algebraic-order dominant terms. This proves (10.6). $\square$

10.1 An explicit all-orders prescription

Write $R(t) = \sum_{j \geq 0} R_j t^j$ in (10.7). For every fixed integer $J \geq 0$, singularity transfer gives

$$a_n = -\lambda \rho^{-n} \sum_{j=0}^J R_j \frac{\Gamma(n-j-\frac{1}{2})}{\Gamma(-j-\frac{1}{2})\Gamma(n+1)} + O(\rho^{-n} n^{-J-5/2}). \quad (10.10)$$

Thus the coefficients of the complete dominant inverse-power expansion are obtained by differentiating the explicit analytic function R and expanding elementary gamma ratios. This prescription avoids an auxiliary unknown recurrence for the asymptotic coefficients. It also makes the error order at every finite truncation explicit.

At $r = 2$, the computations supplied with this report give

$$\begin{aligned} \rho &\approx 0.32542163481800348746, \\ \rho^{-1} &\approx 3.07293644001039978597, \\ C &\approx 5.05502055022293177233, \\ c_1 &\approx -4.11576700299955384839. \end{aligned} \quad (10.11)$$

The table compares exact coefficients with the first two approximations:

n	$a_n/(C\rho^{-n}n^{-3/2})$	$a_n/(C\rho^{-n}n^{-3/2}(1+c_1/n))$
25	0.8421704141	1.0081414221
50	0.9207369642	1.0033260927
100	0.9597064447	1.0009012063
200	0.9796476789	1.0002312733
400	0.9897684471	1.0000584662

The numerical script uses 80-digit decimal arithmetic for the constants and integer arithmetic for the coefficients. These values illustrate the proved asymptotics; they are not used in the Somos proof.

11 Reproducibility and audit trail

The archive is organized so that the exact proof checks can be run without installing a computer algebra system. From its root directory, execute

```
python3 code/verify_certificates.py --audit
python3 code/examples.py
```

The first command recomputes all certificate polynomials and the rank witnesses, then performs the 858 integer audits. The second regenerates sequence files and the asymptotic comparison data. An optional independent symbolic check is

```
python3 code/cross_check_sympy.py
```

which requires SymPy. The standard-library verifier requires Python 3.10 or later and makes no network calls.

The main certificate file records integer coefficients in ascending powers of r . For example, $[1, -2, 3]$ denotes $1 - 2r + 3r^2$, and the zero polynomial is $[0]$. Its moment array is indexed from 0 through 20; its Hankel array is indexed from -1 through 10. This convention is explicitly stored and used by both verifiers, avoiding silent off-by-one shifts.

The delivered files have the following roles:

File or directory	Purpose
article.tex, article.pdf	The mathematical report and its editable source.
code/verify_certificates.py	Exact integer-polynomial determinant proof checks.
code/cross_check_sympy.py	Independent optional symbolic audit.
code/examples.py	Exact moment/Hankel data and decimal asymptotic comparisons.
data/certificates.json	All finite polynomial certificates and rank witnesses.
data/numeric_audit.json	The scope and result of the 858 auxiliary checks.
data/C12_r*_*.txt	Indexed moment and Hankel sequences at five integer parameters.
data/asymptotics_C12_r2.json	High-precision constants and comparison ratios.
README.md, build.sh	Usage, proof dependencies, and compilation instructions.

The TeX source compiles with pdfLaTeX. The build script runs it twice to resolve references and the table of contents. No font files or copies of third-party papers are redistributed. The bibliography links to the original sources.

12 Scope, significance, and remaining questions

The main outcome is a proof of three precise published Somos–8 assertions, not merely the discovery of matching initial terms. The decisive observation is that the universal genus-two theorem should be used as a *polynomial rank bound*, rather than by substituting into its potentially singular rational coordinate formulas. This both validates the degenerate $c_2 = 0$ specialization and makes a small family-specific certificate sufficient.

The work also separates three issues that can otherwise be conflated. Integrality follows from the moment construction. A bilinear identity follows from the rank theorem and exact coefficient identification. A division-based recurrence additionally requires nonzero divisors. Our theorem settles the first two without silently assuming the third.

The exceptional values $r = 1, 3$ in the second family are not discarded; they yield explicit order-six relations. The positive tree formula and the asymptotic expansion add information about the underlying moment sequence. They do not provide Hankel determinant asymptotics, which would require further analysis of the relevant genus-two dynamics and its degenerations.

Several natural directions remain: classifying all rank drops and all zero patterns at specialized parameters; obtaining similarly compact certificates for other low-degree algebraic moment families; developing a direct family-specific determinant or continued-fraction proof that does not import Hone’s general theorem; and deriving asymptotics for the determinant sequences themselves. These directions are not asserted to be solved here.

Finally, the novelty qualification remains important. The theorem and its certificates are fully specified and independently reproducible, but an unlocated prior application of Hone’s theorem would change the historical claim, not the validity of these identities. External mathematical review would be appropriate before treating the report as a publication-ready priority claim.

A The other two rank witnesses

For reference, the matrices formed from rows 7, 8, 9, 10 at $r = 2$ for the other two families are

$$M_{11}(2) = \begin{pmatrix} -135120 & -34593 & -1430 & 1681 \\ 14703657 & -270240 & 472771 & 511225 \\ 1376443582 & -200949979 & 96610800 & 132963961 \\ -2390410472409 & -492078580565 & -56515956289 & 18257414400 \end{pmatrix}, \quad (\text{A.1})$$

and

$$M_{13}(2) = \begin{pmatrix} -1733 & -1186 & -78 & 169 \\ 57264 & -1733 & 7709 & 6084 \\ 1058325 & -372216 & 135174 & 351649 \\ -259181611 & -82549350 & -16978776 & 3003289 \end{pmatrix}. \quad (\text{A.2})$$

Their exact determinants are listed in (7.2). The witness calculation does not claim that $M_k(r)$ is invertible for every r . It proves invertibility over the rational function field, which is all the finite-certification lemma needs before polynomial specialization.

B A compact logical checklist

The proof can be audited in the following order.

1. Confirm the three continued fractions and recurrence parameters in Barry’s Conjectures 11–13, with the stated correction to the separate Catalan display for Conjecture 13.
2. Verify the compact moment recursion (3.2) by multiplying out (2.4).
3. Verify the universal moment form (4.1) and its match to Hone’s Theorem 5.4 via (4.6).

4. Pass to polynomial minors before setting $c_2 = 0$, using Lemma 4.2.
5. Check the binomial translation and the seven substituted parameters in (5.10).
6. Recompute the twelve zero residual polynomials and the three nonzero rank witnesses.
7. Apply Lemma 6.1, then specialize the resulting $\mathbb{Z}[r]$ -identities, without dividing by a Hankel determinant.

Every family-specific claim needed for the all-index result is covered by these steps. The auxiliary numerical tests and the asymptotic section are not premises of the main theorem.

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